

SANJAY DESHPANDE

✉ sanjay.deshpande1@northwestern.edu 🌐 www.sanjaydeshpande.phd
in [linkedin.com/in/sanjaydeshpandeph](https://www.linkedin.com/in/sanjaydeshpandeph)
📍 Trenton, New Jersey

🎓 EDUCATION

Yale University <i>Doctor of Philosophy in Electrical Engineering (GPA: 4.0)</i>	<i>New Haven, CT</i> <i>Aug. 2021 – Dec 2025</i>
George Mason University <i>Master of Science in Computer Engineering (GPA: 3.83)</i>	<i>Fairfax, VA</i> <i>Aug. 2014 – Dec 2016</i>
Jawaharlal Nehru Technological University <i>Bachelor of Technology in Electronics & Communication Engineering (GPA: 4.0)</i>	<i>Hyderabad, India</i> <i>Sep. 2010 – May 2014</i>

💼 WORK EXPERIENCE

Postdoctoral Scholar <i>Northwestern University</i>	<i>Jun 2025 - current</i> <i>Remote</i>
- Researching side-channel attacks on post-quantum cryptographic algorithms and developing countermeasures. - Designing hardware accelerators for fully homomorphic encryption schemes. - Designing architectures for quantum circuit integrity verification in cloud-based quantum computers. - Research contributions leading to peer-reviewed publications and open-source implementations.	
PQC Research Scientist Resident <i>SandboxAQ</i>	<i>Apr 2023 - Dec 2023 & Nov 2024 - May 2025</i> <i>Remote</i>
- Evaluated performance of submissions from NIST's Post-Quantum Digital Signature Schemes standardization competition. - Implemented hardware design of Syndrome Decoding in the Head (SDitH) signature scheme from the NIST post-quantum digital signature standardization process. - Designed and implemented modular RTL architecture for the NIST-proposed Rijndael 256 algorithm. - Work resulted in a publication at IACR TCHES and publicly released open-source implementations.	
Research Intern <i>Microsoft Research</i>	<i>Jun 2024 - Aug 2024</i> <i>Redmond, WA</i>
- Designed a modular RTL architecture for an ISO-standardized post-quantum cryptographic scheme targeting ASIC, supporting runtime switching between security levels and synthesis-time reconfigurability for performance trade-offs. - Extended the design with power side-channel attack protections. - Work resulted in a publication at IACR TCHES and publicly released open-source implementations.	
Research Assistant / Associate Researcher <i>Yale University</i>	<i>Oct 2020 - May 2025</i> <i>New Haven, CT</i>
- Designed and implemented efficient RTL for post-quantum key encapsulation and digital signature algorithms across multiple NIST PQC competition candidates, targeting both FPGA and ASIC. - Developed hardware implementations of key components in public-key and post-quantum cryptography, optimizing for area, timing, and power - Proposed and implemented novel software and hardware solutions for securing cloud-based quantum computing platforms. - Designed RISC-V-compatible hardware accelerators for cryptographic workloads. - Research resulted in 15+ peer-reviewed publications at top venues (IACR TCHES, IEEE HOST, SAC, ACM TECS) and multiple open-source implementations.	
Senior Security Researcher <i>Technology Innovation Institute (formerly DarkMatter LLC)</i>	<i>Mar 2017 -Jul 2020</i> <i>Abu Dhabi, UAE</i>
- Developed platform-independent RTL implementations of cryptographic algorithms and protocols for both FPGA and ASIC targets, from requirements through production-ready IP. - Implemented hardware accelerators for elliptic curve cryptography and post-quantum cryptography primitives.	

- Optimized RTL designs for performance in terms of Power, Timing, Frequency, and Area. Drew test plans for verification of the IP.
- Optimized designs for power, timing, frequency, and area; drew test plans for verification and validation.
- Analyzed RTL implementations for side-channel vulnerabilities and developed resistant countermeasures.
- Conducted hardware reverse engineering, hacking, and security threat assessment on various devices.

Research Assistant

Aug 2015 – Dec 2016

Cryptographic Engineering Research Group (CERG), George Mason University

Fairfax, VA

- Implemented cryptographic algorithms in VHDL and Verilog targeting Xilinx Virtex 6, Virtex 7, and Zynq 7000 platforms.
- Identified performance bottlenecks of authenticated ciphers on hardware and designed methods to overcome them.
- Work resulted in a publication at Euromicro DSD.

Junior Electrical Engineer

May 2015 – Aug 2015

Rysc Corp.

Manassas, VA

- Designed and prototyped electronic hardware; conducted PCB design (Eagle CAD) and developed ARM firmware.

Lab Assistant

Aug 2014 – Aug 2015

George Mason University

Fairfax, VA

- Assisted students in experiments based on Electrical and Computer Engineering.
- Verified the quality of the components used in Lab Experiments.

TEACHING EXPERIENCE

Yale University

New Haven, CT

Teaching Fellow

- CPSC415/CPSC515: Law, Security, and Logic
- EENG201: Introduction to Computer Engineering

Fall 2022
Spring 2023, Spring 2024

George Mason University

Fairfax, VA

Teaching Assistant

- ECE334: Linear Electronics Lab
- ECE545: Digital System Design with VHDL
- ECE448: FPGA Design with VHDL Lab
- ECE232: Digital System Design Lab

Fall 2015, Fall 2016
Fall 2015, Fall 2016
Spring 2016
Summer 2016

ACADEMIC ACTIVITIES

Visiting Scholar

Nov 2025

Academia Sinica

Taipei, Taiwan

- Built a methodology that standardizes hardware evaluation practices and ensures consistent design assessment.

Academic Reviewing Service

- IEEE Symposium on Security and Privacy 2026
- International Symposium on Circuits and Systems (ISCAS) 2026
- Journal of Systems Architecture
- IEEE Transactions on Computers
- CHIPS Journal
- IEEE Computer Architecture Letters
- International Symposium on Circuits and Systems (ISCAS) 2024

Mar 2026
Nov 2025
Oct 2025
Oct 2025
Jul 2025
Jun 2024
Dec 2023

Program Committee Member

- IACR Transactions on Cryptographic Hardware and Embedded Systems (TCHES) 2027 *Jul 2026*
- International Conference on Security, Privacy and Applied Cryptographic Engineering (SPACE) 2026 *Jun 2026*
- IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS) 2026 *Dec 2025*
- Hardware and Architectural Support for Security and Privacy (HASP) 2025 *Sept 2025*
- ACM QSec: Quantum Security and Privacy Workshop (Co-located with ACM CCS 2025) *Oct 2025*

Session Chair

- 2nd IEEE Conference on Secure and Trustworthy Cyber Infrastructure for IoT and Microelectronics (SaTC 2026) *Mar 2026*

Invited Talks

- Cedarcrypt 2026, Paphos, Cyprus *Jul 2026*
- PQSecure Technologies, Boca Raton, USA *Apr 2026*
- Microsoft Research, Redmond, USA *Jan 2026*
- Academia Sinica, Taipei, Taiwan *Nov 2025*
- New York University, New York, USA *Oct 2025*
- Indian Institute of Technology, Kanpur, India *Aug 2025*
- Department of Electrical and Computer Engineering, Northwestern University, Evanston, USA *May 2025*
- Security Group, Qualcomm, Valbonne, France *Nov 2023*

Podcasts

- Boardwalk Bytes, Atlantic City, New Jersey, USA *Jul 2025*
- SandboxAQ, Palo Alto, California, USA *Jun 2024*

Awards

- Best Hardware Demo (Second Place) award, IEEE HOST 2026 *May 2026*
- Best of IEEE CAL award for a paper on Quantum Computer Security, IEEE CAL *Dec 2023*
- Best Project Award, Cyber-physical Systems Summer School, Sardegna, Italy *Sept 2025*
- Outstanding Academic Achievement Award, ECE Department at GMU, Fairfax, USA *May 2015*
- Best project award in Cryptography and Computer Network Security course at GMU, Fairfax, USA. *Jul 2014*
- Certificate of Excellence in Academics, Jawaharlal Nehru Technological University, India *Jul 2014*
- First Prize in Robocup 2k14, a zonal event, Jawaharlal Nehru Technological University, India *Oct 2013*

PEER-REVIEWED PUBLICATIONS AND RESEARCH

Cryptography and Hardware Research

1. Stef Halmans, Niklas Höher, Dina Hesse, **Sanjay Deshpande**, Jakub Szefer, and Tim Güneysu. Storing less in-the-head: An area-efficient hardware architecture for SDitH-v2. *Cryptology ePrint Archive*, Paper 2026/938, 2026
2. **Sanjay Deshpande**, Patrick Longa, and Jakub Szefer. Accelerating FrodoKEM in Hardware. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, 2026(3), Oct. 2026
3. Nishant Pandey, **Sanjay Deshpande**, Dixit Dutt Bohra, Debapriya Basu Roy, Dip Sankar Banerjee, and Jakub Szefer. Performance analysis of parameterizable HQC hardware architecture. In *Proceedings of the Selected Areas in Cryptography*, SAC, Aug. 2026
4. **Sanjay Deshpande**, Yongseok Lee, Mamuri Nawan, Kashif Nawaz, Ruben Niederhagen, Yunheung Paek, and Jakub Szefer. Unified MEDS Accelerator. *SAC*, August 2025
5. **Sanjay Deshpande**, Yongseok Lee, Cansu Karakuzu, Jakub Szefer, and Yunheung Paek. SPHINCSLET: An Area-Efficient Accelerator for the Full SPHINCS+ Digital Signature Algorithm. *ACM Trans. Embed. Comput. Syst.*, April 2025
6. **Sanjay Deshpande**, James Howe, Dongze Yue, and Jakub Szefer. SDitH in Hardware. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, 2023(2), Sept. 2024
7. **Sanjay Deshpande**, Chuanqi Xu, Mamuri Nawan, Kashif Nawaz, and Jakub Szefer. Fast and Efficient Hardware Implementation of HQC. In *Proceedings of the Selected Areas in Cryptography*, SAC, Aug. 2023
8. Po-Jen Chen, Tung Chou, **Sanjay Deshpande**, Norman Lahr, Ruben Niederhagen, Jakub Szefer, and Wen Wang. Complete and improved FPGA Implementation of Classic McEliece. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, 2022(3):71–113, Jun. 2022
9. **Sanjay Deshpande**, Santos Merino del Pozo, Victor Mateu, Marc Manzano, Najwa Aaraj, and Jakub Szefer. Modular Inverse for Integers using Fast Constant Time GCD Algorithm and its Applications. In *Proceedings of the International Conference on Field Programmable Logic and Applications*, FPL, Aug. 2021
10. **Sanjay Deshpande** and Kris Gaj. Analysis and Inner-Round Pipelined Implementation of Selected Parallelizable CAESAR Competition Candidates. In *2017 Euromicro Conference on Digital System Design (DSD)*, pages 274–282, 2017

Quantum Computer Security Research

11. Barbora Hrdá, **Sanjay Deshpande**, Theodoros Trochatos, and Jakub Szefer. CHEQ: Towards Enabling Circuit Integrity Checking in Quantum Controllers. In *Proceedings of the Great Lakes Symposium on VLSI 2025*, GLSVLSI, pages 265–272, June 2025
12. Theodoros Trochatos, Chuanqi Xu, **Sanjay Deshpande**, Yao Lu, Yongshan Ding, and Jakub Szefer. Trusted Execution Environments for Quantum Computers. *Frontiers in Computer Science*, 7, 2025
13. Theodoros Trochatos, Chuanqi Xu, **Sanjay Deshpande**, Yao Lu, Yongshan Ding, and Jakub Szefer. Protecting Quantum Computers with a Trusted Controller. In *International Conference on Quantum Computing and Engineering*, QCE, September 2024
14. Theodoros Trochatos, **Sanjay Deshpande**, Chuanqi Xu, Yao Lu, Yongshan Ding, and Jakub Szefer. Dynamic Pulse Switching for Protection of Quantum Computation on Untrusted Clouds. In *International Symposium on Hardware Oriented Security and Trust*, HOST, May 2024
15. Theodoros Trochatos, Chuanqi Xu, **Sanjay Deshpande**, Yao Lu, Yongshan Ding, and Jakub Szefer. A Quantum Computer Trusted Execution Environment. *IEEE Computer Architecture Letters*, (01):1–4, Oct. 2023
16. **Sanjay Deshpande**, Chuanqi Xu, Theodoros Trochatos, Hanrui Wang, Ferhat Erata, Song Han, Yongshan Ding, and Jakub Szefer. Design of Quantum Computer Antivirus. In *Proceedings of the International Symposium on Hardware Oriented Security and Trust*, HOST, May 2023
17. **Sanjay Deshpande**, Chuanqi Xu, Theodoros Trochatos, Yongshan Ding, and Jakub Szefer. Towards an Antivirus for Quantum Computers. In *Proceedings of the International Symposium on Hardware Oriented Security and Trust*, HOST, Jun. 2022

Other Research

18. **Sanjay Deshpande** and Jakub Szefer. Analyzing ChatGPT’s Aptitude in an Introductory Computer Engineering Course. In *Proceedings of the International Conference on Frontiers in Education: Computer Science & Computer Engineering*, FECS, Jul. 2023

Demos

19. Huizhen Zhou, Luke Beckwith, **Sanjay Deshpande**, Xiaokuan Zhang, Kris Gaj, Jakub Szefer. Correlation Power Analysis of Hamming-Quasi Cyclic. In *International Symposium on Hardware Oriented Security and Trust*, HOST, May 2026
20. Theodoros Trochatos, **Sanjay Deshpande**, and Jakub Szefer. SoteriaQ: Securing Quantum Circuits Hardware Demo. In *International Symposium on Hardware Oriented Security and Trust*, HOST, May 2024

Book Chapters

21. **Deshpande, Sanjay**, Chuanqi Xu, Theodoros Trochatos, Hanrui Wang, Ferhat Erata, Song Han, Yongshan Ding, and Jakub Szefer. *Design of Quantum Computer Antivirus*, pages 373–398. Springer Nature Switzerland, Cham, 2026
22. Theodoros Trochatos, Chuanqi Xu, **Deshpande, Sanjay**, Yao Lu, Yongshan Ding, and Jakub Szefer. *Trusted Execution Environments for Quantum Computers*, pages 399–446. Springer Nature Switzerland, Cham, 2026

PhD Dissertation

23. **Deshpande, Sanjay**. *Hardware Accelerators for Post-Quantum Cryptography – A Comprehensive Study of PQC Hardware Accelerators Belonging to Different Families*. PhD thesis, Yale University, 2025. ProQuest Dissertations & Theses (Publication No. 31934449)

PROJECTS

Optimized Decomposition of $U_{Heis3}(t)$ into Quantum Gates | *Qiskit, Python, IBMQ*

Mar 2022 – May 2022

Participated in the IBM challenge and contributed to improving the fidelity of the Heisenberg Hamiltonian H_{Heis3} circuit. Implemented a ‘**trotter**’ function using optimal two-qubit transformations and demonstrated a fidelity of 52% on **ibmq-jakarta**.

Proposed a method to detect malicious circuits in quantum programs. Explored the possibility of modifying Qiskit, and added multi-layered protection – an Antivirus system to detect the malicious attacker circuits, which would prevent malicious users from performing attacks, proposed to run Qiskit programs inside a trusted execution environment, protecting it from malicious users.

Survey on Inference Acceleration for various NN models on cloud processors

Oct 2021 – Dec 2021

Benchmarked results for inference acceleration of pre-trained DNN models – ResNet50 and MobileNetV1 on cloud FPGAs, cloud GPUs, and cloud CPUs. Built a common framework to track different performance metrics - Time, Accuracy, Throughput, and Energy. Provided a fair performance comparison for DNN inference based on the different performance metrics on cloud CPU, cloud GPU, and cloud FPGA.

Complete ASIC Design Flow using Synopsys ASIC design tools | *ASIC, Verilog*

Aug 2015 – Dec 2015

Implemented an ALU and carried out the complete ASIC design flow – used Design Compiler for floorplanning, place, and route, and PrimeTime for clock tree insertion and power estimation, generated area, power, and timing reports, and located the critical paths of the designs. Optimized false paths and maximum delay paths. Used IC Compiler to create back-end designs and generated the GDSII files.

High-Level Synthesis of an ALU | *FPGA, High Level Synthesis*

Jan 2015 – May 2015

Designed an ALU using high-level synthesis, created a custom IP to Vivado, and interfaced it with Zynq 7000 using AXI Lite and AXI Stream interfaces.

LANGUAGES

- English
- Hindi
- Telugu
- Marathi
- Kannada

REFERENCES

Prof. Jakub Szefer

jakub.szefer@northwestern.edu

Northwestern University

Evanston, IL, USA

Prof. Krzysztof M. Gaj

kgaj@gmu.edu

George Mason University

Fairfax, VA, USA

Prof. Ruzica Piskac

ruzica.piskac@yale.edu

Yale University

New Haven, CT, USA

Prof. Brandon Reagen

bjr5@nyu.edu

New York University

New York, NY, USA

Prof. Ruben Niederhagen

ruben@iis.sinica.edu.tw

Academia Sinica

Taipei, Taiwan